



Letter Ballot

Mplify 141.1

LSO Cantata and LSO Sonata Billing Management API - Developer Guide

July 2025

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List of Contributing Members

The following members of Mplify participated in the development of this document and have requested to be included in this list.

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Table 1. Contributing Members

1. Abstract

This standard is intended to assist implementation of the Billing functionality defined for the LSO Cantata and LSO Sonata Interface Reference Points (IRPs), for which requirements and use cases are defined in MEF 134 *Billing and Invoice Business Requirements and Use Cases* [MEF 134]. This standard consists of this document and complementary API definitions for:

- Billing Management and Billing Notification.

This standard normatively incorporates the following files by reference as if they were part of this document from the GitHub repository

MEF-LSO-Sonata-SDK

commit id: [aaa03d484f98664a5a14f4f54f47b675d7efb3b8](#)

- [productApi/billing/billingManagement.api.yaml](#)
- [productApi/billing/billingNotification.api.yaml](#)

MEF-LSO-Cantata-SDK

commit id: [83d6edd0c70386058a9af6e677c069b498671da7](#)

- [productApi/billing/billingManagement.api.yaml](#)
- [productApi/billing/billingNotification.api.yaml](#)

The Billing API is defined using OpenAPI 3.0 [OAS-V3]

2. Terminology and Abbreviations

This section defines the terms used in this document. In many cases, the normative definitions of terms are found in other documents. In these cases, the third column is used to provide the reference that is controlling, in other Mplify or external documents.

In addition, terms defined in the standards referenced below are included in this document by reference and are not repeated in Table 2:

- MEF 55.1
- MEF 55.1.1
- Mplify 150

Term	Description	Reference
Application Program Interface (API)	In the context of LSO, API describes one of the Management Interface Reference Points based on the requirements specified in an Interface Profile, along with a data model, the protocol that defines operations on the data and the encoding format used to encode data according to the data model. In this document, API is used synonymously with REST API.	[MEF 55.1]
Bill	A legal document generated by the Seller to the Buyer relating to charges associated to Products provided by the Seller to the Buyer.	[MEF 134]
Buyer	In the context of this document, denotes the organization or individual acting as the customer in a transaction over a Cantata (Customer <-> Service Provider) or Sonata (Service Provider <-> Partner) Interface.	This document; adapted from [MEF 55.1.1]
Bill Item	One or more rows in a Bill that represent charges associated with a Product instance.	[MEF 134]
Invoice	A legal document generated by the Seller to the Buyer relating to charges associated to Products provided by the Seller to the Buyer. Within this document, an Invoice term is referred to as Bill	[MEF 134]
Notification	A message sent from the Seller to the Buyer to inform about an event that has occurred in regard to a specific instance of Billing	[MEF 134]
Printable Bill	An Bill that is in a format that can be printed and reviewed by a human.	[MEF 134]
Requesting Entity	The business organization that is acting on behalf of one or more Buyers. In the most common case, the Requesting Entity represents only one Buyer and these terms are then synonymous.	[Mplify 150]
Responding Entity	The business organization that is acting on behalf of one or more Sellers. In the most common case, the Responding Entity represents only one Seller and these terms are then synonymous.	[Mplify 150]
REST API	Representational State Transfer. REST provides a set of architectural constraints that, when applied as a whole, emphasizes the scalability of component interactions, the generality of interfaces, the independent deployment of components, and intermediary components to reduce interaction latency, enforce security, and encapsulate legacy systems.	[REST]

Seller	In the context of this document, denotes the organization acting as the supplier in a transaction over a Cantata (Customer <-> Service Provider) or Sonata (Service Provider <-> Partner) Interface.	This document; adapted from [MEF 55.1.1]
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Table 2. Terminology

3. Compliance Levels

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 (RFC 2119 [[RFC2119](#)], RFC 8174 [[RFC8174](#)]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as **[Ox]** for optional.

A paragraph preceded by **[CRa]<** specifies a conditional mandatory requirement that **MUST** be followed if the condition(s) following the "<" have been met. For example, "**[CR1]<[D38]**" indicates that Conditional Mandatory Requirement 1 must be followed if Desirable Requirement 38 has been met. A paragraph preceded by **[CDb]<** specifies a Conditional Desirable Requirement that **SHOULD** be followed if the condition(s) following the "<" have been met. A paragraph preceded by ****[COc]<*** specifies a Conditional Optional Requirement that **MAY** be followed if the condition(s) following the "<" have been met.

4. Introduction

The Billing API allows the Buyer to search or retrieve Bills as well as receive notifications.

This standard specification document describes the Application Programming Interface (API) for Billing functionality of the LSO Cantata and LSO Sonata Interface Reference Point (IRP) as defined in the *MEF 55.1 Lifecycle Service Orchestration (LSO): Reference Architecture and Framework* [MEF 55.1]. The LSO Reference Architecture is shown in Figure 1 with both IRPs highlighted.

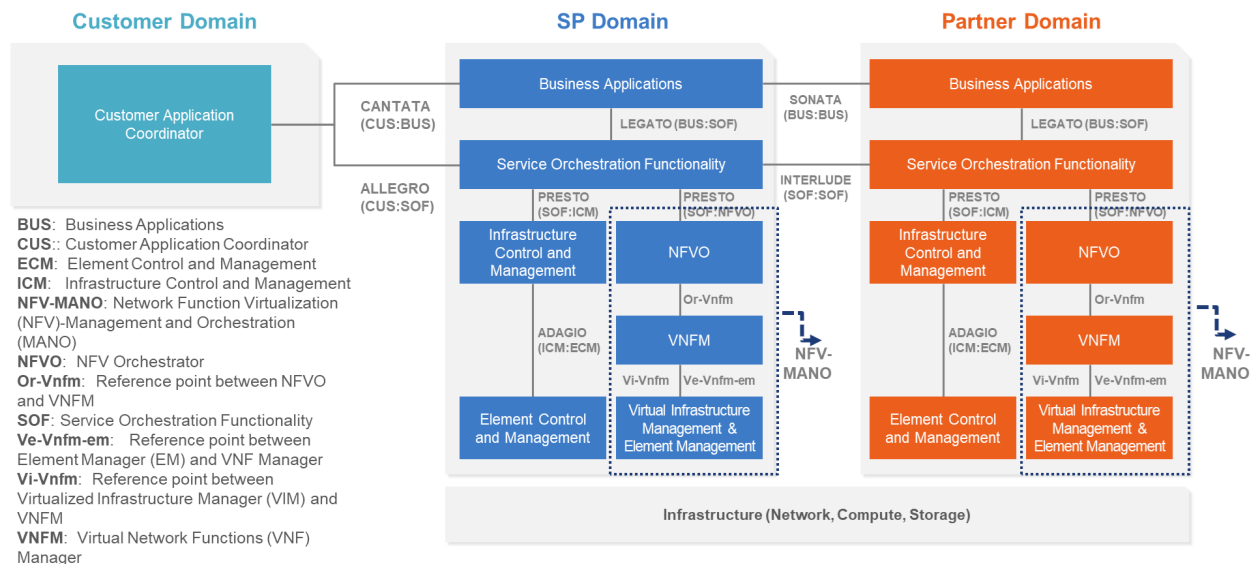


Figure 1. The LSO Reference Architecture

Cantata and Sonata IRPs define functionalities that allow an automated exchange of information between business applications of the Buyer (Customer or Service Provider) and Seller (Service Provider or Partner) Domains. Those are:

- Product Catalog
- Address Validation
- Site Retrieval
- Product Offering Qualification
- Product Quote
- Product Inventory
- Product Ordering
- Trouble Ticketing
- Billing
- Appointment
- WorkOrder

This API and Developer Guide implements requirements and use cases for Billing as defined in MEF 134 *Billing and Invoice Business Requirements and Use Cases* [MEF 134].

This document is structured as follows:

- [Chapter 4](#) provides an introduction to Billing description in a broader context of Cantata and Sonata and their corresponding SDKs.
- [Chapter 5](#) gives an overview of endpoints, resource models, and design patterns.
- Use Cases and flows are presented in [Chapter 6](#).
- And finally, [Chapter 7](#) complements previous sections with a detailed API description.

4.1. Conventions in the Document

- Code samples are formatted using code blocks. When notation `<< some text >>` is used in the payload sample it indicates that a comment is provided instead of an example value and it might not comply with the OpenAPI definition.
- Model definitions are formatted as in-line code (e.g. `Billing`).
- In UML diagrams the default cardinality of associations is `0..1`. Other cardinality markers are compliant with the UML standard.
- In the API details tables and UML diagrams required attributes are marked with a `*` next to their names.
- In UML sequence diagrams `{{variable}}` notation is used to indicate a variable to be substituted with a correct value.

4.2. Relation to Other Documents

This API implements the Billing requirements and use cases that are defined in [MEF134]. The API definition builds on *TMF678 Customer Bill Management API User Guide v4.0.1* [TMF678]. In the context of naming, the terms Invoice and Bill are used alternatively. The Invoice term appears in [MEF 134] document, the Bill term is used in the context of this document and the API to reuse the terms of TMF 678 API.

4.3. Approach

As presented in Figure 2 both Cantata and Sonata API frameworks consist of three structural components:

- Generic API framework
- Product-independent information (Function-specific information and Function-specific operations)
- Product-specific information (Mplify product specification data model)

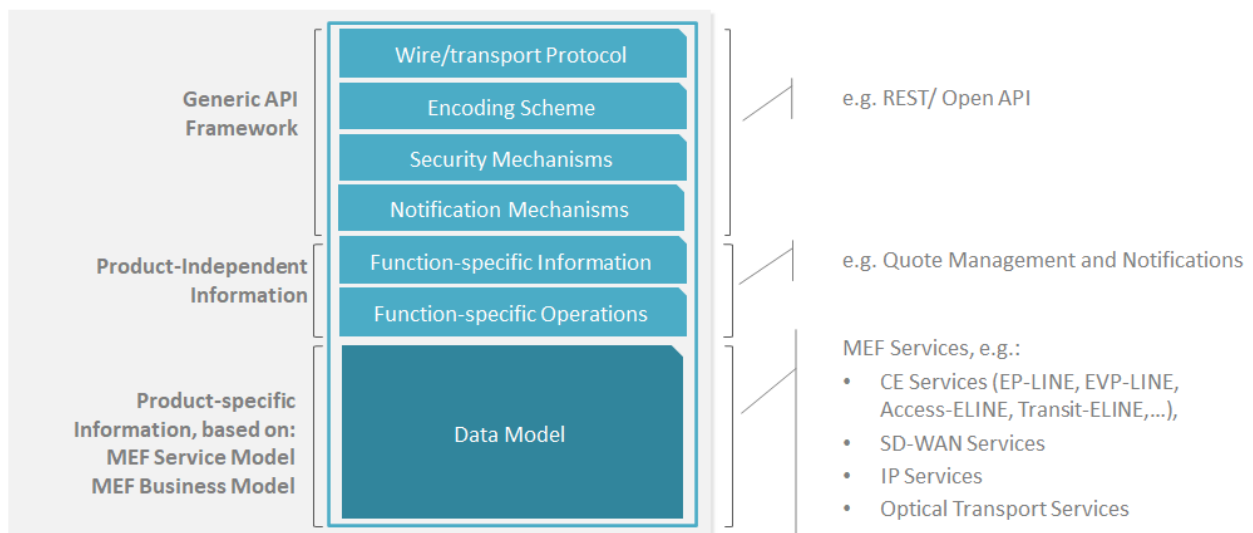


Figure 2. Cantata and Sonata API framework

The essential concept behind the framework is to decouple the common structure, information and operations from the specific product information content.

Firstly, the Generic API Framework defines a set of design rules and patterns that are applied across all Cantata or Sonata APIs.

Secondly, the product-independent information of the framework focuses on a model of a particular Cantata or Sonata functionality and is agnostic to any of the product specifications.

Finally, the product-specific information part of the framework focuses on Mplify product specifications that define business-relevant attributes and requirements for trading Mplify subscriber and Mplify operator services.

The Billing is product-agnostic and is not intended to carry any product-specific payloads. It only references products from the inventory by **id**. It operates using the Generic API Framework and the Function-specific Information and Operations.

4.4. High-Level Flow

The Billing is part of a broader Cantata and Sonata End-to-End flow. Figure 3. below shows a high-level diagram to get a good understanding of the whole process and the Billing position within it.

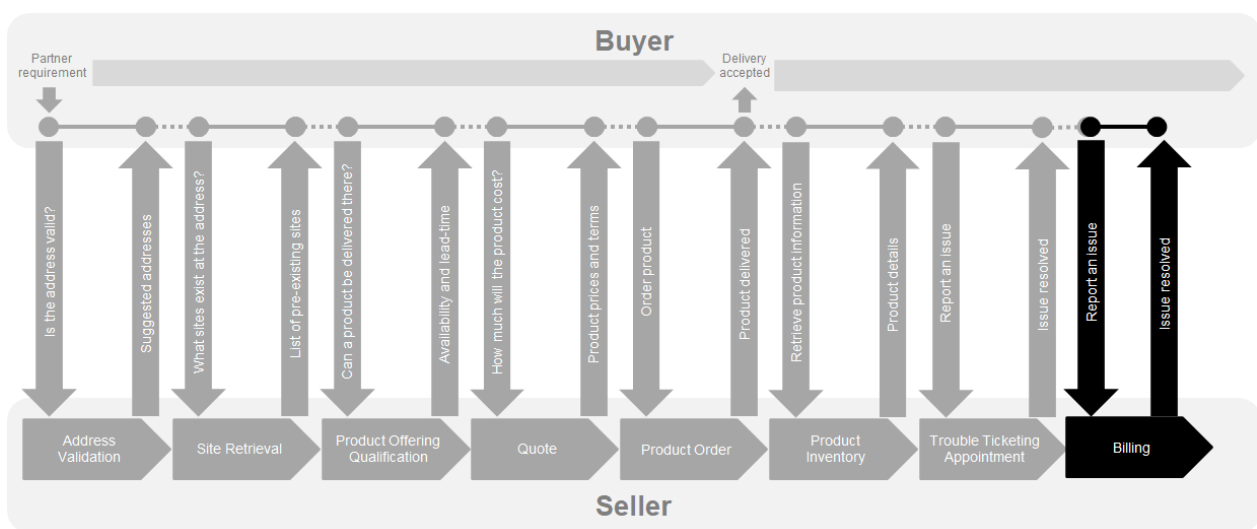


Figure 3. Cantata and Sonata End-to-End Function Flow

- Address Validation:
 - Allows the Buyer to retrieve address information from the Seller, including exact formats, for addresses known to the Seller.
- Site Retrieval:
 - Allows the Buyer to retrieve Geographic Site information including exact formats for Geographic Sites known to the Seller.
- Product Offering Qualification (POQ):
 - Allows the Buyer to check whether the Seller can deliver a product or set of products from among their product offerings at the geographic address or a Geographic Site specified by the Buyer; or modify a previously purchased product.
- Quote:
 - Allows the Buyer to submit a request to find out how much the installation of an instance of a Product Offering, an update to an existing Product, or a disconnect of an existing Product will cost.
- Product Order:

- Allows the Buyer to request the Seller to initiate and complete the fulfillment process of an installation of a Product Offering, an update to an existing Product, or a disconnect of an existing Product at the address defined by the Buyer.
- Product Inventory:
 - Allows the Buyer to retrieve information about existing Product instances from Seller's Product Inventory.
- Billing:
 - Allows the Seller to generate the document to the Buyer relating to charges associated with Products provided by the Seller to the Buyer.
- Trouble Ticketing:
 - Allows the Buyer to create, retrieve, and update Trouble Tickets as well as receive notifications about Incidents' and Trouble Tickets' updates. This allows managing issues and situations for a Product provided by the Seller.
- The Appointment and WorkOrder:
 - Allows the Buyer to create, retrieve, and update Appointment and WorkOrder as well as receive notifications about Appointment and WorkOrder' updates.

5. API Description

This section presents the API structure and design patterns. It starts with the high-level use cases diagram. Then it describes the REST endpoints with use case mapping. Next, it gives an overview of the API resource model.

5.1. High-Level Use Cases

Figure 4 presents a high-level use case diagram as specified in MEF 134 [MEF134] in section 7. This picture aims to help understand the endpoint mapping. Use Cases are described extensively in chapter 6.

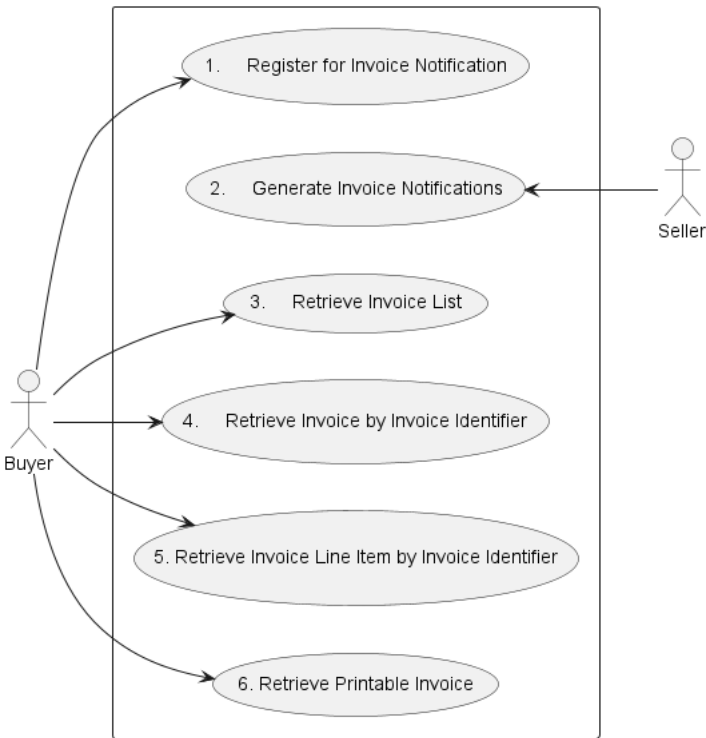


Figure 4. Use Cases

5.2. API Endpoint and Operation Description

5.2.1. Seller-side API Endpoints

Base URL for Cantata: `https://{serverBase}/mefApi/cantata/customerBillManagement/v3/`

Base URL for Sonata: `https://{serverBase}/mefApi/sonata/customerBillManagement/v3/`

The following API endpoints are implemented by the Seller and allow the Buyer to retrieve Customer Bills and register for Notifications. The endpoints and corresponding data model are defined in:

`/productApi/billing/billingManagement.api.yaml`

API endpoint	Description	MEF 134 Use Case mapping
<code>POST /hub</code>	Buyer is able to register for Customer Bill Notifications.	UC 1: Register for Invoice Notifications

API endpoint	Description	MEF 134 Use Case mapping
GET /hub/{id}}	Buyer is able to register for Customer Bill Notifications.	UC 1. Register for Invoice Notifications
DELETE /hub/{id}}	Buyer is able to register for Customer Bill Notifications.	UC 1. Register for Invoice Notifications

Table 3. Seller-side optional API endpoints

[O1] The Seller implementation **MAY** support API endpoints listed in Table 3. [MEF134 R2]

API endpoint	Description	MEF 134 Use Case mapping
GET /customerBill	The Buyer is able to retrieve a list of Customer Bills.	UC 3: Retrieve Invoice List
GET /customerBill/{id}}	The Buyer is able to retrieve a specific Customer Bill based on the identifier.	UC 4: Retrieve Invoice by Identifier
GET /customerBillItem/{id}}	The Buyer is able to retrieve a specific Invoice based on the Identifier.	UC 5: Retrieve Line Item by Identifier

Table 4. Seller-side mandatory API endpoints

[R1] The Seller implementation **MUST** support API endpoints listed in Table 4.

5.2.2. Buyer-side API Endpoints

Base URL for Cantata: <https://{serverBase}/mefApi/cantata/customerBillNotification/v3/>

Base URL for Sonata: <https://{serverBase}/mefApi/sonata/customerBillNotification/v3/>

The following API endpoints are implemented by the Buyer and allow the Seller to send Notifications. The endpoints and corresponding data model are defined in:

[/productApi/billing/billingNotification.api.yaml](#).

API endpoint	Description	MEF 134 Use Case mapping
POST /listener/customerBillCreateEvent	Seller sends a Notification to a Buyer.	UC 2. Generate Invoice Notification
POST /listener/customerBillStateChangeEvent	Seller sends a Notification to a Buyer.	UC 2. Generate Invoice Notification

Table 5. Buyer-side optional Billing API endpoints

[O2] The Buyer implementation **MAY** support API endpoints listed in Table 5. [MEF134 R2]

5.3. Specifying the Buyer ID and the Seller ID

A business entity willing to represent multiple Buyers or multiple Sellers must follow requirements of [Mplify 150] chapter 8.8, which states:

For requests of all types, there is a business entity that is initiating an Operation (called a Requesting Entity) and a business entity that is responding to this request (called the

Responding Entity). In the simplest case, the Requesting Entity is the Buyer, and the Responding Entity is the Seller. However, in some cases, the Requesting Entity may represent more than one Buyer and similarly, the Responding Entity may represent more than one Seller.

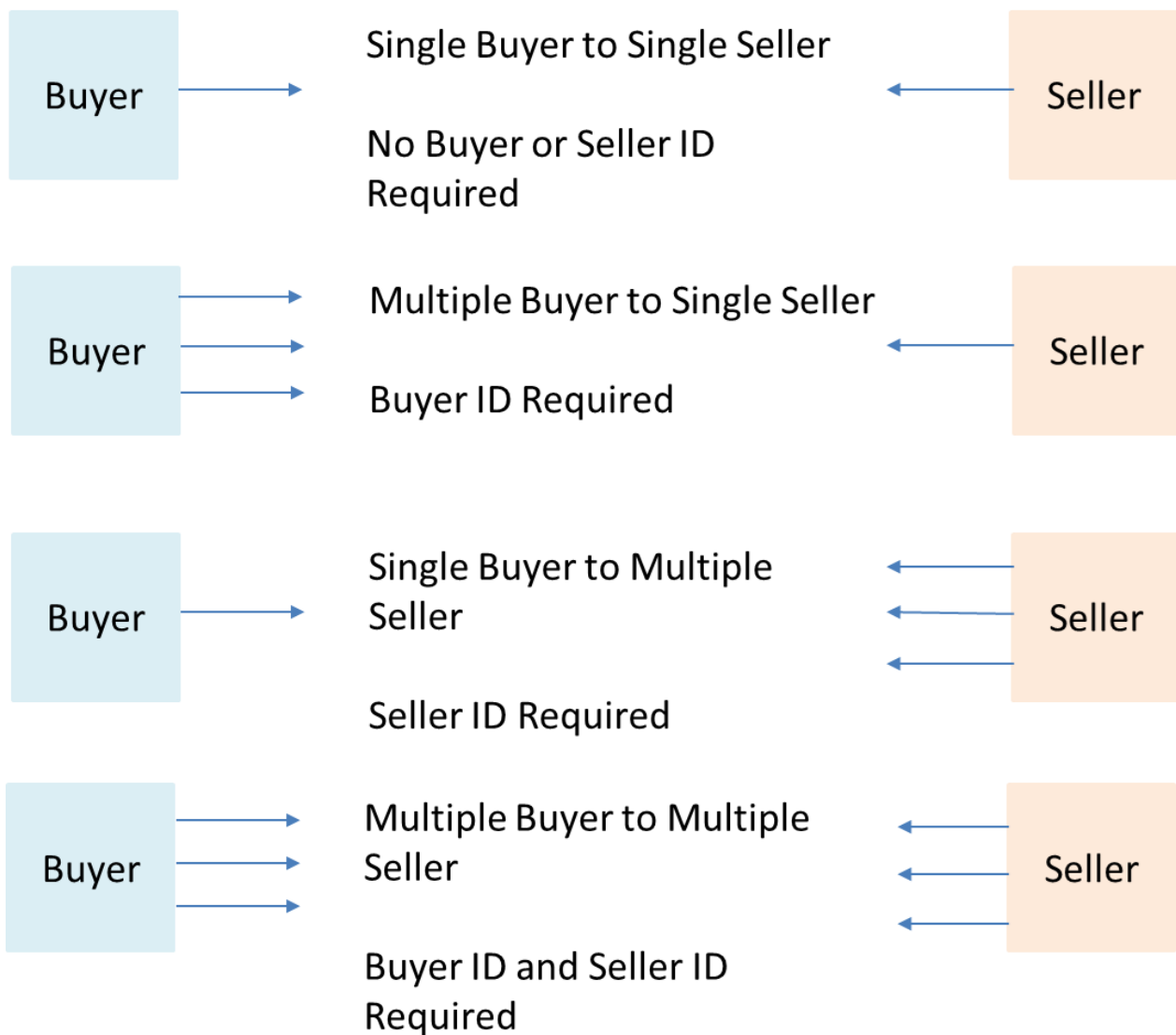


Figure 5. Buyer ID and Seller ID Examples

As shown in Figure 5, if a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing a single Seller, Buyer and Seller IDs are not required to be passed between the two entities. If a Requesting Entity representing more than one Buyer is doing business with a Responding Entity representing a single Seller, the Buyer ID is required to be passed between the two entities. If a Requesting Entity representing a single Buyer is doing business with a Responding entity representing multiple Sellers, the Seller ID is required to be passed between the two entities. If a Requesting Entity representing multiple Buyers is doing business with a Responding Entity representing multiple Sellers, both the Buyer ID and the Seller ID are required to be passed between the entities.

While it is outside the scope of this specification, it is assumed that the Requesting Entity and the Responding Entity are aware of each other and can authenticate requests initiated by the other party. It is further assumed that the Requesting Entity knows:

- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity; and

- the list of Sellers that this Responding Entity represents to this Requesting Entity.

It is also assumed that the Responding Entity knows:

- the list of Sellers that this Responding Entity represents to this Requesting Entity and
- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity.

In the API the **buyerId** and **sellerId** are represented as optional query parameters in each operation defined.

[R2] If the Requesting Entity has the authority to represent more than one Buyer the request **MUST** include **buyerId** that identifies the Buyer being represented. [Mplify150 R62]

[R3] If the Responding Entity represents more than one Seller to this Buyer the request **MUST** include **sellerId** that identifies the Seller with whom this request is associated. [Mplify150 R63]

5.4. Model Structural Validation

The structure of the HTTP payloads exchanged via Trouble Ticket API endpoints is defined using OpenAPI version 3.0.

[R4] Implementations **MUST** use payloads that conform to these definitions.

[R5] The Buyer and the Seller **MUST NOT** use any operation, entity or attribute that is not explicitly defined or allowed by this standard.

5.5. Security Considerations

There must be an authentication mechanism whereby a Seller can be assured who a Buyer is and vice-versa. There must also be authorization mechanisms in place to control what a particular Buyer or Seller is allowed to do and what information may be obtained. However, the definition of the exact security mechanism and configuration is outside the scope of this document. Security considerations are standardized by *LSO API Security Profile* [MEF 128.1].

6. API Interactions and Flows

This section provides a detailed insight into the API functionality, use cases, and flows. It starts with Figure 6 and Table 6, presenting a list and short description of all business use cases then presents the variants of end-to-end interaction flows, and the following subchapters describe the API usage flow and examples for each of the use cases.

Figure 6 presents an example of an end-to-end flow:

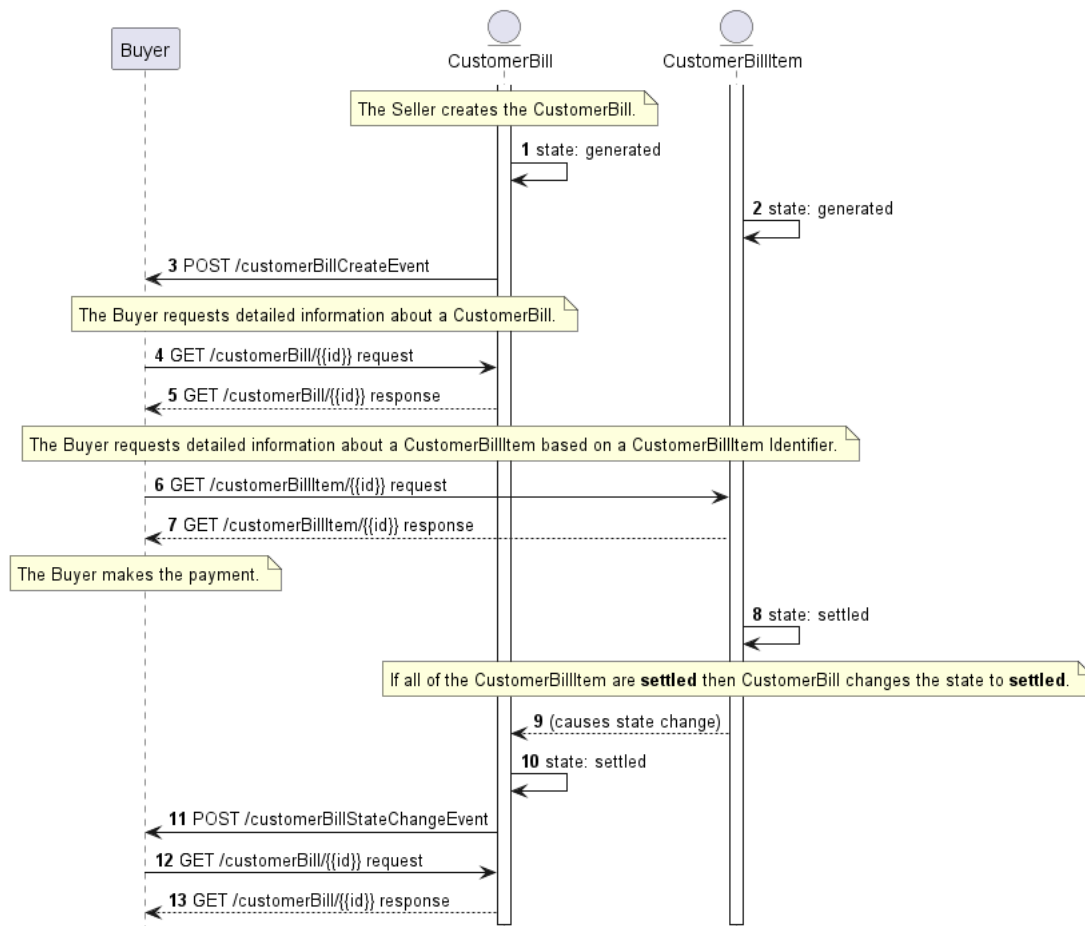


Figure 6. End-to-End API Flows

- (1,2) The Seller creates a **CustomerBill** and **CustomerBillItem** in the **generated** state.
- (3) The Seller sends a **customerBillCreateEvent** with an **id**.
- (4,5) The Buyer requests detailed information about the **CustomerBill** based on **id**.
- (6,7) The Buyer requests detailed information about all of **CustomerBillItem** based on the list of **ids** from **CustomerBill**.
- (8) The Buyer pays the bill and **CustomerBillItem** changes state to **settled**.
- (9,10) If all of the **CustomerBillItem** are **settled** then **CustomerBill** changes the state to **settled**.
- (11) The Seller sends a **customerBillStateChangeEvent**.
- (12,13) The Buyer requests detailed information about the **CustomerBill** based on **id**.

Use

Case #	Use Case Name	Use Case Description
1	Register for Invoice Notification	Buyer is able to register for Invoice Notifications.

Use Case #	Use Case Name	Use Case Description
2	Generate Invoice Notifications	Seller sends an Invoice Notification to a Buyer.
3	Retrieve Invoice List	The Buyer is able to retrieve a list of Invoices.
4	Retrieve Invoice by Invoice Identifier	The Buyer is able to retrieve a specific Invoice based on the Invoice Identifier of the Invoice.
5	Retrieve Invoice Line Item by Invoice Identifier	The Buyer is able to retrieve one or more specific Line Items in an Invoice.
6	Retrieve Printable Invoice	The Buyer is able to retrieve a specific Invoice in a printable format using the Invoice Identifier of the Invoice.

Table 6. Use Cases description

The detailed business requirements of each of the use cases are described in section 7 of MEF 134 [MEF 134].

6.1. Use Case 1: Register for Invoice Notifications

To register for notifications the Buyer uses the `registerListener` operation from the API: `POST /hub`.

- `callback` - mandatory, to provide the callback address the events will be notified to,
- `query` - optional, to provide the required types of event.

[R6] The Seller **MUST** support all of `CustomerBillEventType`: [MEF134 R1]

- `customerBillCreateEvent`
- `customerBillStateChangeEvent`

[R7] The Buyer's request **MUST** provide the `callback` attribute. [MEF134 R1]

By using a simple request:

```
{
  "callback": "https://buyer.mef.com/listenerEndpoint"
}
```

The Buyer subscribes for notification of all types of events.

If the Buyer wishes to receive only notification of a certain type, a `query` must be added:

```
{
  "callback": "https://buyer.mef.com/listenerEndpoint",
  "query": "eventType=customerBillCreateEvent"
}
```

If the Buyer wishes to subscribe to different types of events, there are 2 possible syntax variants [TMF630]:

```
eventType=customerBillCreateEvent,customerBillStateChangeEvent
```

or

```
eventType=customerBillCreateEvent&eventType=customerBillStateChangeEvent
```

The **query** formatting complies with RCF3986 [RFC3986](#). According to it, every attribute defined in the Event model (from the notification API) can be used in the **query**. However, this standard requires only **eventType** attribute to be supported.

[R8] If the Seller does not support notifications, they **MUST** return an error message to a Buyer's request to register for notifications that indicates that notifications are not supported (**Error501**). [MEF134 R2]

[R9] **eventType** is the only attribute that the Seller **MUST** support in the query.

The Seller responds to the subscription request by adding the **id** of the subscription to the message that must be further used for unsubscribing.

```
{
  "id": "00000000-0000-0000-0000-000000000678",
  "callback": "https://buyer.mef.com/listenerEndpoint",
  "query": "eventType=customerBillCreateEvent"
}
```

Example of a final address that the Notifications will be sent to (for Sonata, **customerBillCreateEvent**):

- <https://buyer.mef.com/listenerEndpoint/mefApi/sonata/customerBillManagement/v3/listener/customerBillCreateEvent>

6.2. Use Case 2: Send Invoice Notification

Notifications are used to asynchronously inform the Buyer about the respective objects and attributes changes. The next notification must be sent when the state changes compared to the previously sent one.

[R10] The event sent by the Seller **MUST** contain following attributes: [MEF134 R3]

- **eventId**
- **eventType**
- **eventTime**
- **event:**
 - **id**
 - **state** (in case of **CustomerBillStateChangeEvent**)

[R11] The **customerBillCreateEvent** **MUST** be sent when a **CustomerBill** is created. [MEF134 R4]

[R12] The Seller **MUST** send Customer Bill Notifications to a Buyer who has registered for the Customer Bill Notification Type.

[R13] The Seller **MUST NOT** send Customer Bill Notifications for Notification Type to a Buyer who has not registered for the Customer Bill Notification Type.

The Buyer acknowledges the Notification received from the Seller.

[O3] If the Seller fails to receive an acknowledgment from the Buyer repeatedly, the Seller **MAY** make the target address as bad and stop sending notifications. [MEF134 O1]

Figure 7 shows all entities involved in the Notification use cases.

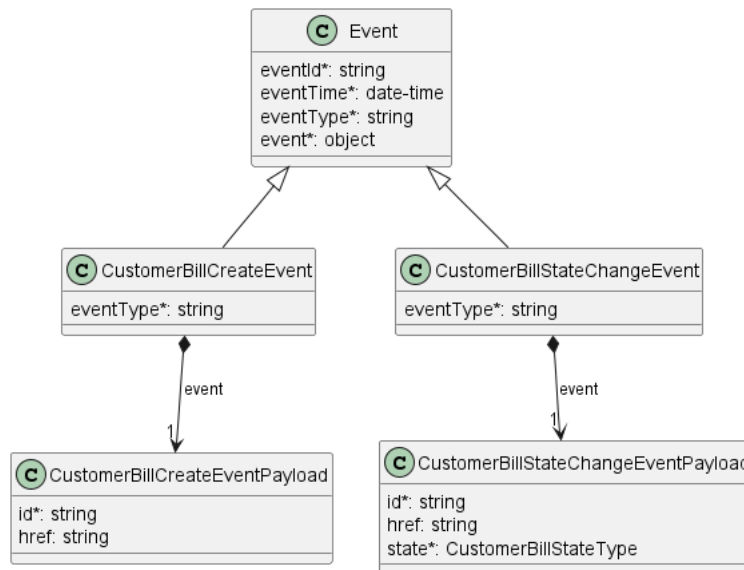


Figure 7. Use Case 2. Billing Notification Data Model

The following snippet presents an example of `customerBillCreateEvent`

```

{
  "eventId": "event-001",
  "eventType": "customerBillCreateEvent",
  "eventTime": "2023-05-09T15:56:08.559Z",
  "event": {
    "id": "00000000-4444-5555-6666-000000000987"
  }
}
  
```

Note: The body of the event carries only the source object's `id`. The Buyer needs to query it later by `id` to get details.

To stop receiving events, the Buyer has to use the `unregisterListener` operation from the `DELETE /hub/{id}` endpoint. The `id` is the identifier received from the Seller during the listener registration.

Table 7 presents the mapping between the API Notification types' names and the ones in MEF 134. The inconsistencies are caused by using the TMF event types as the base for this API.

API name	MEF 134 name
<code>customerBillCreateEvent</code>	Created
<code>customerBillStateChangeEvent</code>	State Change

Table 7. Customer Bill event types

6.3. Use Case 3: Retrieve Invoice List

The Buyer can get detailed information about the Bill from the Seller by using a **GET** `/customerBill` operation.

[R14] The Buyer request **MUST** contain zero or more of the following query parameters: [MEF134 R5]

- `billingAccount.id`
- `billingPeriod.startDateTime.lt`
- `billingPeriod.startDateTime.gt`
- `billingPeriod.endDateTime.lt`
- `billingPeriod.endDateTime.gt`
- `category`
- `state`

The Seller receives this request and returns a response.

[R15] The Seller response **MUST** include the following for each entry in the list: [MEF134 R6]

- `id`
- `billNo`
- `billingAccount`
- `billingPeriod`
- `category`.
- `state`

The example below shows a response for use case 3.

```
[
  {
    "id": "CB-123",
    "billingAccount": {
      "id": "00000000-1111-0000-0000-000000000001"
    },
    "billNo": "780123456",
    "billingPeriod": {
      "startDateTime": "2022-10-01T08:00:00.297Z",
      "endDateTime": "2022-10-31T08:00:00.297Z"
    },
    "category": "normal",
    "state": "generated"
  }
]
```

[R16] If no **CustomerBill** matches the criteria provided by the Buyer, the Seller **MUST** return a positive response with an empty list. [MEF134 R7]

The Buyer may also ask for pagination of the response when the number of results is too big. The following query attributes related to pagination can be provided:

- `limit` - number of expected list items
- `offset` - offset of the first element in the result list.

```
https://serverRoot/mefApi/sonata/customerBillManagement/v3/customerBill?state=generated&limit=10&offset=0
```

The example above shows a Buyer's request to get all Bills that are in the **generated** state. Additionally, the Buyer asks only for a first (`offset=0`) pack of 10 results (`limit=10`) to be returned. The correct response (HTTP code **200**) in the response body contains a list of

`CustomerBill_Find` objects matching the criteria. To get more details (e.g. the item level information), the Buyer has to query a specific `CustomerBill` by `id`.

The Seller returns a list of elements that comply with the requested `limit`. If the requested `limit` is higher than the supported list size then the smaller list of results is returned. In that case, the size of the result is returned in the header attribute `X-Result-Count`. The Seller can indicate that there are additional results available using:

- `X-Total-Count` header attribute with the total number of available results
- `X-Pagination-Throttled` header set to `true`

[D1] The Seller **SHOULD** support the pagination mechanism.

[CR1]<[D1] Seller **MUST** use either `X-Total-Count` or `X-Pagination-Throttled` to indicate that the page was truncated and additional results are available.

6.4. Use Case 4: Retrieve Invoice by Identifier

The Buyer can get detailed information about the Bill from the Seller by using a `GET /customerBill/{id}` operation.

The Seller receives this request and returns a response.

[R17] The Seller response **MUST** include all of the attributes for the returned `CustomerBill`.
[MEF134 R9]

- `id`
- `amountDue`
- `appliedPayment`
- `billingAccount`
- `billCycle`
- `billDate`
- `billDocument`
- `billNo`
- `billingPeriod`
- `category`
- `credits`
- `customerBillItem`
- `discounts`
- `fees`
- `financialAccount`
- `lastUpdate`
- `paymentDueDate`
- `runType`
- `relatedContactInformation`
- `remainingAmount`
- `state`
- `taxExcludedAmount`
- `taxIncludedAmount`
- `taxItem`

The example below shows a response for use case 4.

```
[
  {
    "id": "CB-123",
```

```

    "amountDue": {<< The total amount of money that needs to be paid by Buyer, with all taxes, fees, and credits
    applied. >>
      "unit": "EUR",
      "value": 120.0
    },
    "appliedPayment": [],<<No payments made yet>
    "billingAccount": {<<The Buyer Billing Account that is unique within the Seller and is assigned by the Seller>>
      "id": "00000000-1111-0000-0000-000000000001"
    },
    "billCycle": "BC-1234",<<The Bill cycle as set by the Seller>>
    "billDate": "2022-09-31T10:30:00.846Z",<<The date when the Bill was issued.>>
    "billDocument": {<<URL pointing to PDF file, which is used by the Buyer to receive the Bill>>
      "url": "https://example.com/documents/00000000-0000-1111-2222-000000001111"
    },
    "billNo": "780123456",<<A number assigned to the Bill by the Seller>>
    "billingPeriod": {<<The time, when service will be delivered.>>
      "startDateTime": "2022-10-01T08:00:00.297Z",
      "endDateTime": "2022-10-31T08:00:00.297Z"
    },
    "category": "normal",<<The category of Bill>>
    "customerBillItem": [<<A reference to one or more Bill Items.>>
      {
        "id": "ABR123"
      },
      {
        "id": "ABR124"
      }
    ],
    "credits": { <<Amount of credits included>>
      "unit": "EUR",
      "value": 0.0
    },
    "discounts": {<<Amount of discounts included>>
      "unit": "EUR",
      "value": 0.0
    },
    "fees": {<<Amount of fees included>>
      "unit": "EUR",
      "value": 10.0
    },
    "financialAccount": {<<A Financial Account within the Seller.>>
      "id": "23-0000-0000-3324-3332-3334"
    },
    "lastUpdate": "2022-09-31T10:30:00.846Z",<<The date when the Bill was last modified e.g. date of changing the
    state of Bill>>
    "paymentDueDate": "2022-10-31T08:00:00.846Z",<<Final payment date.>>
    "runType": "onCycle",<<The Bill cycle as set by the Seller>>
    "relatedContactInformation": [<<Detailed contact information about Buyer. >>
      {
        "emailAddress": "john.example@example.com",
        "name": "John Example",
        "number": "+12-345-678-90",
        "organization": "Buyer Example Co.",
        "role": "buyerBillingContact"
      }
    ],
    "remainingAmount": {<<An amount of money that still requires payment e. g. from the previous Bill Period.>>
      "unit": "EUR",
      "value": 120.0
    },
    "state": "generated",<<The state of the Bill>>
    "taxExcludedAmount": {<<The amount of money due without taxes being calculated >>
      "unit": "EUR",
      "value": 100.0
    },
    "taxIncludedAmount": {<<The amount of money due with taxes being calculated for the current Invoicing Cycle.>>
      "unit": "EUR",
      "value": 120.0
    },
    "taxItem": [<<The tax items including category, rate, and amount for this Bill.>>
      {
        "taxCategory": "VAT",
        "taxRate": 20.0,
        "taxAmount": {
          "unit": "EUR",
          "value": 20.0
        }
      }
    ]
  }
}

```


1. This example describes the case when Bill is in a **generated** state. It means that the Buyer needs to pay it.
2. The **appliedPayment** is an empty list and the **remainingAmount** value is "120", which is equal to **amountDue**.
3. When the Buyer pays the full bill amount, the **CustomerBill** will change the state to **settled**.
4. A record describing the received payment is added to the **appliedPayment** list. The received **appliedAmount** (120) is deducted from **remainingAmount**.

```
[
  {
    "id": "CB-123",
    "amountDue": {<< Does not change - it is the total amount>>
      "unit": "EUR",
      "value": 120.0
    },
    "appliedPayment": [ <<Now reflecting received payemtns>>
      {
        "appliedAmount": {<<The amount of money that was received from the Buyer and applied to the Bill as payment
without additional fees.>>
          "unit": "EUR",
          "value": 120.0
        },
        "payment": {
          "id": "PAY-9876",
          "amount": {<<The amount of money received from Buyer.>>
            "unit": "EUR",
            "value": 120.0
          },
          "paymentMethod": "electronic",<<The specific method of payment. >>
          "paymentDate": "2022-10-25T09:00:00.846Z"<<The Date the payment was received by the Seller.>
        }
      }
    ],
    "billingAccount": {
      "id": "00000000-1111-0000-0000-000000000001"
    },
    "billCycle": "BC-1234",
    "billDate": "2022-09-31T10:30:00.846Z",
    "billDocument": {
      "url": "https://example.com/documents/00000000-0000-1111-2222-000000001111"
    },
    "billNo": "780123456",
    "billingPeriod": {
      "startDateTime": "2022-10-01T08:00:00.297Z",
      "endDateTime": "2022-10-31T08:00:00.297Z"
    },
    "category": "normal",
    "customerBillItem": [
      {
        "id": "ABR123"
      },
      {
        "id": "ABR124"
      }
    ],
    "credits": {
      "unit": "EUR",
      "value": 0.0
    },
    "discounts": {
      "unit": "EUR",
      "value": 0.0
    },
    "fees": {
      "unit": "EUR",
      "value": 10.0
    },
    "financialAccount": {
      "id": "23-0000-0000-3324-3332-3334"
    },
    "lastUpdate": "2022-10-25T09:00:00.846Z",<<Changed to date of receiving the payment due to state change>>
    "paymentDueDate": "2022-10-31T08:00:00.846Z",
    "runType": "onCycle",
  }
]
```

```

"relatedContactInformation": [
  {
    "emailAddress": "john.example@example.com",
    "name": "John Example",
    "number": "+12-345-678-90",
    "organization": "Buyer Example Co.",
    "role": "buyerBillingContact"
  }
],
"remainingAmount": {<<Now 0.0 as the payments were received>>
  "unit": "EUR",
  "value": 0.0
},
"state": "settled",<<Settled - all payments received>
"taxExcludedAmount": {
  "unit": "EUR",
  "value": 100.0
},
"taxIncludedAmount": {
  "unit": "EUR",
  "value": 130.0
},
"taxItem": [
  {
    "taxCategory": "VAT",
    "taxRate": 20.0,
    "taxAmount": {
      "unit": "EUR",
      "value": 20.0
    }
  }
]
}
]

```

6.4.1. CustomerBill - Lifecycle

Figure 8 presents the CustomerBill state machine:

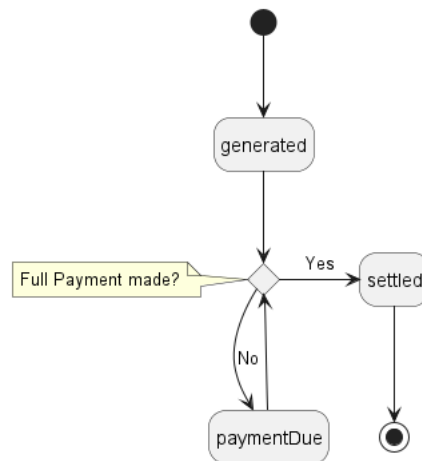


Figure 8. CustomerBill State Machine

1. A **CustomerBill** has been **generated** based on the completion of a CustomerBill Cycle and a **CustomerBill** Notification has been sent to the Buyer.
2. In the **paymentDue** state, the Seller has completed their dispute process and they are expecting payment from the Buyer.
3. In a **settled** state a **CustomerBill** for which all payment due has been received or for which payment is due.

Table 8 presents the mapping between the API **state** names (aligned with TMF) and the MEF 134 naming, together with the states' descriptions.

state	MEF 134 name	Description
generated	GENERATED	An CustomerBill that has been generated based on the completion of a CustomerBill Cycle and a CustomerBill Notification has been sent to the Buyer.
paymentDue	PAYMENT_DUE	The Seller has completed their dispute process and they are expecting payment from the Buyer.
settled	SETTLED	A CustomerBill for which all payment due has been received or for which payment is due.

Table 8. Customer Bill states

[R18] A CustomerBill API implementation **MUST** support the states and transitions shown in Figure 8. [MEF134 R19]

6.5. Use Case 5: Retrieve Invoice Line Item by Identifier

The Buyer can get detailed information about the Bill from the Seller by using a **GET /customerBillItem/{id}** operation.

The Seller receives this request and returns a response.

[R19] The Seller response **MUST** include the attributes for the returned **CustomerBillItem**: [MEF134 R10], [MEF134 R11], [MEF134 R13]

- **id**
- **appliedTax**
- **appliedFee**
- **description**
- **periodCoverage**
- **product** or **productOrderItem**
- **productName**
- **state**
- **taxExcludedAmount**
- **type**
- **unit**
- **unitRate**
- **unitQuantity**

[R20] If the bill item relates to an instance of a Product then the **product** reference **MUST** be provided. [MEF134 R10]

[R21] If the bill item relates to a Product Order Item then the **productOrderItem** reference **MUST** be provided. [MEF134 R11]

The first example shows below a response of one **customerBillItem** in use case 5. This example shows the **customerBillItem** attributes before payment.

```
[
  {
    "id": "ABR123", <<The Bill Item identifier.>>
    "appliedTax": [<<Taxes associated with the Bill Item>>
      {
        "category": "country",
        "rate": 20, <<The rate at which the Tax is calculated>>
        "description": "Country Tax",
        "amount": { <<The amount of money of the Tax. e. g. 20% of 50 gives 10 >>
          "unit": "EUR",
```

```

        "value": 10.0
    }
}
],
"appliedFee": [<<Fees associated with the Bill Item>>
{
    "category": "recurring",
    "rate": 10,<<The rate at which the Fee is calculated>>
    "description": "Recurring Fee",
    "amount": {
        "unit": "EUR",<<The amount of money of the Fee. e. g. 10% of 70 gives 14 >>
        "value": 5.0
    }
}
],
"customerBillItemType": "recurring",
"description": "Subscriber Operator charge",<<A text description of the charge.>>
"periodCoverage": {<<The time, when service related to Bill Item will be delivered>>
    "startDateTime": "2022-10-01T08:00:00.297Z",
    "endDateTime": "2022-10-31T08:00:00.297Z"
},
"product": {<<The identifier of the Product that is the subject of the Bill Item.>>
    "id": "EVLAN1345"
},
"productOrderItem": {
    "productOrderId": "00000000-5555-0000-0000-000000000022",
    "productOrderItemId": "item-001"
},

"productName": "Elan_connectivity", <<The name of the Product that is the subject of the Line Item.>>
"state": "generated",
"taxExcludedAmount": { <<The amount of money of the charge before taxes and fees are calculated and applied,
e.g. net.>>
    "unit": "EUR",
    "value": 50
},
"unit": "month",
"unitRate": { <<The rate per unit for the Bill determined during or after the Billing Process.>>
    "unit": "EUR",
    "value": 65
},
"unitQuantity": 1 <<The number of units.>>
}
]

```

The second example shows below a response of another one **customerBillItem** in use case 5. This example shows the **customerBillItem** attributes after payment.

```

[
{
    "id": "ABR124",
    "appliedTax": [
        {
            "category": "country",
            "rate": 20,
            "description": "Country Tax",
            "amount": {
                "unit": "EUR",
                "value": 10.0
            }
        }
    ],
    "appliedFee": [
        {
            "category": "recurring",
            "rate": 10,
            "description": "Recurring Fee",
            "amount": {
                "unit": "EUR",
                "value": 5.0
            }
        }
    ],
    "customerBillItemType": "recurring",
    "description": "Subscriber Operator charge",
    "periodCoverage": {

```

```

    "startDateTime": "2022-10-01T08:00:00.297Z",
    "endDateTime": "2022-10-31T08:00:00.297Z"
  },
  "product": {
    "id": "ELAN1345"
  },
  "productOrderItem": {
    "productOrderId": "00000000-5555-0000-0000-000000000001",
    "productOrderItemId": "item-002"
  },
  "productName": "Evlan_connectivity",
  "state": "settled",
  "taxExcludedAmount": {
    "unit": "EUR",
    "value": 50
  },
  "unit": "month",
  "unitRate": {
    "unit": "EUR",
    "value": 65
  },
  "unitQuantity": 1
}
]

```

After payment, the **customerBillItem** changes the states to **settled**.

6.5.1. CustomerBillItem - Lifecycle

Figure 9 presents the **CustomerBillItem** state machine:

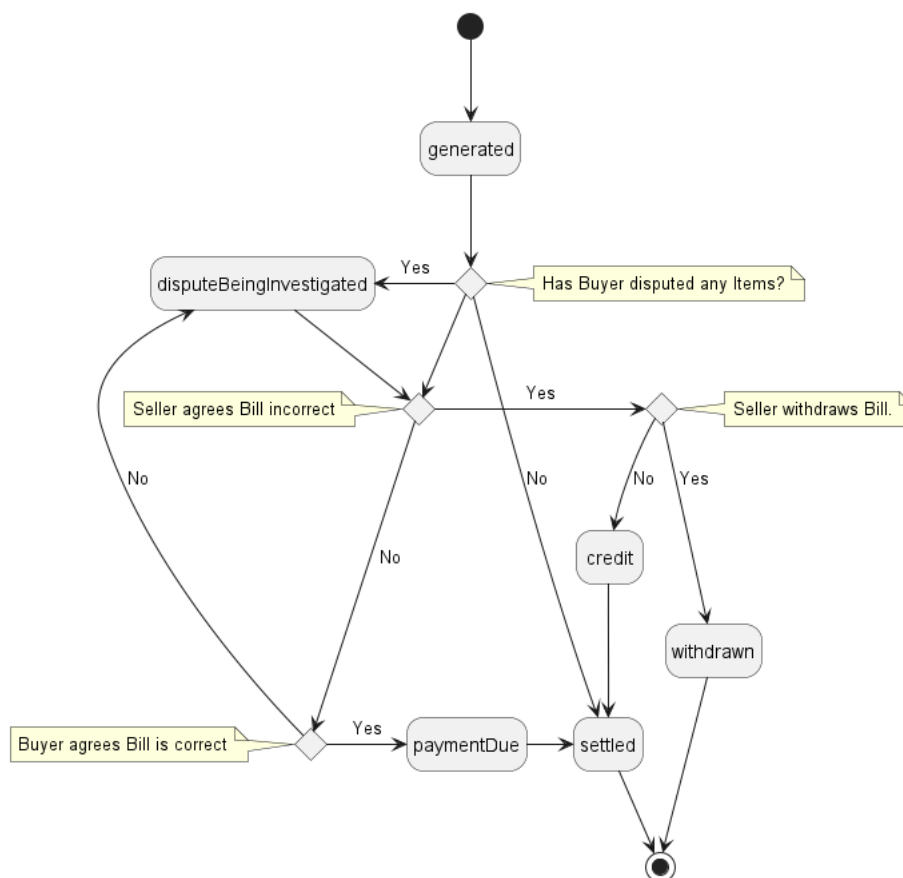


Figure 9. CustomerBillItem State Machine

1. The Seller has agreed with the Buyer that the disputed charges are in error and has provided a **credit** for the amount in error.
2. A Buyer has disputed charges included in a **CustomerBill** and those charges are in the **disputeBeingInvestigated** process. Disputes cannot exit the **disputeBeingInvestigated**

process until the Dispute is resolved.

3. A **CustomerBill** that has been **generated** based on the completion of a Billing Cycle and a **CustomerBill** Notification have been sent to the Buyer.
4. In the **paymentDue** state, the Seller has completed their dispute process and they are expecting payment from the Buyer.
5. The **settled** state means that a **CustomerBill** for which all payments due has been received or payment is due for the **CustomerBillItem**.
6. A Seller has agreed with the Buyer that disputed charges are in error and has **withdrawn** the CustomerBill.

Table 9 presents the mapping between the API **state** names and the MEF 134 naming, together with states' descriptions.

state	MEF 134 name	Description
credit	CREDIT	The Seller has agreed with the Buyer that the disputed charges are in error and has provided a credit for the amount in error.
disputeBeingInvestigated	DISPUTE_BEING_INVESTIGATED	A Buyer has disputed charges included in a CustomerBill and those charges are in the Sellers Dispute Process. Disputes cannot exit the dispute process until the Dispute is resolved.
generated	GENERATED	A CustomerBill that has been created based on the completion of a Billing Cycle and a CustomerBill Notification has been sent to the Buyer.
paymentDue	PAYMENT_DUE	The Seller has completed their dispute process and they are expecting payment from the Buyer.
settled	SETTLED	A CustomerBill for which all payment due has been received or payment is due for the CustomerBillItem .
withdrawn	WITHDRAWN	A Seller has agreed with the Buyer that disputed charges are in error and has withdrawn the CustomerBill.

Table 9. Customer Bill Item states

[R22] A CustomerBill API implementation **MUST** support the states and transitions shown in Figure 9. [MEF134 R20]

The interaction between **CustomerBillItem** and **CustomerBill** states is shown in Figure 10:

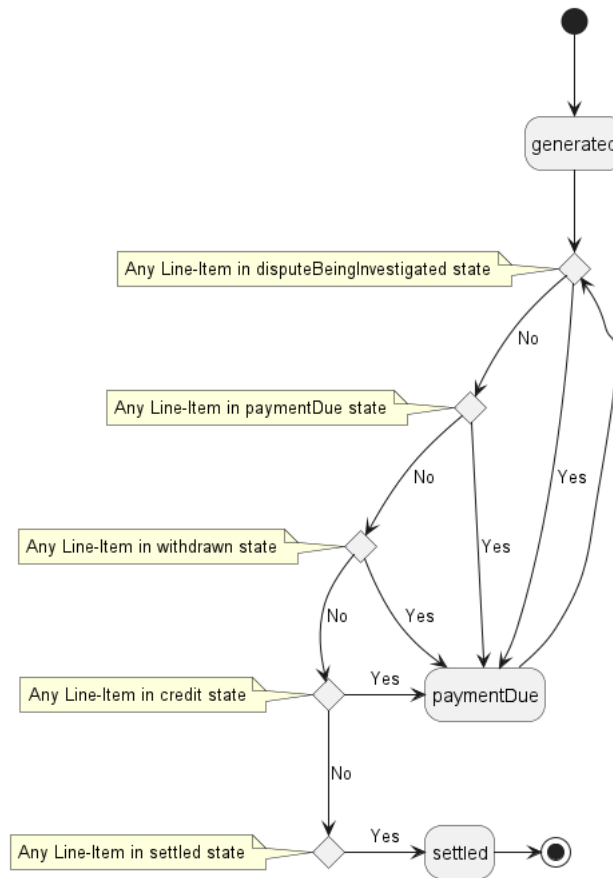


Figure 10. Interaction between CustomerBillItem and CustomerBill States Machine

[R23] A CustomerBill API implementation **MUST** support logic shown in Figure 10. [MEF134 R21]

6.6. Use Case 6: Retrieve Printable Invoice

[R24] If a printable document is agreed upon by the Buyer and Seller, then the Seller **MUST** provide a valid URL in **CustomerBill.billDocument**. [MEF134 R17]

Whether the link point to a pre-generated file or the file is generated dynamically upon request is up the Seller's discretion.

[R25] The Buyer **MUST** use the **billDocument** provided by the Seller to retrieve the Printable CustomerBill. [MEF134 R18]

[R26] The printable **CustomerBill** **MUST** be in a PDF format. [MEF134 R15]

[R27] The printable version of **CustomerBill** **MUST** contain all attributes of **CustomerBill** and **CustomerBillItems**. [MEF134 R16]

7. API Details

7.1. API patterns

7.1.1. Indicating errors

Erroneous situations are indicated by appropriate HTTP responses. An error response is indicated by HTTP status 4xx (for client errors) or 5xx (for server errors) and appropriate response payload. The Product Order API uses the error responses as depicted and described below.

Implementations can use HTTP error codes not specified in this standard in compliance with rules defined in RFC 7231 [RFC7231]. In such a case, the error message body structure might be aligned with the **Error**.

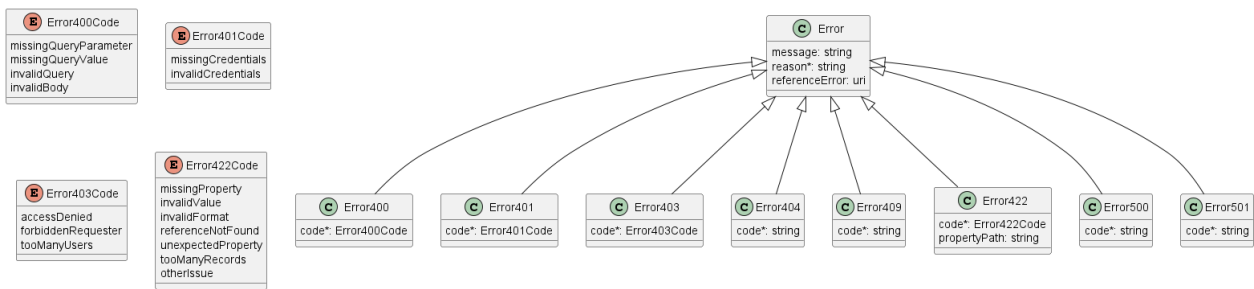


Figure 11. Data model types to represent an erroneous response

7.1.1.1. Type Error

Description: Standard Class used to describe API response error Not intended to be used directly. The **code** in the HTTP header is used as a discriminator for the type of error returned in runtime.

Name	Type	Description
message	string	Text that provides mode details and corrective actions related to the error. This can be shown to a client user.
reason*	string <i>maxLength = 255</i>	Text that explains the reason for the error. This can be shown to a client user.
referenceError	uri <i>format = uri</i>	URL pointing to documentation describing the error

7.1.1.2. Type Error400

Description: Bad Request. (<https://tools.ietf.org/html/rfc7231#section-6.5.1>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

Name	Type	Description
code*	Error400Code	One of the following error codes: - missingQueryParameter: The URI is missing a required query-string parameter - missingQueryValue: The URI is missing a required query-string parameter value - invalidQuery: The query section of the URI is invalid. - invalidBody: The request has an invalid body

7.1.1.3. **enum** Error400Code

Description: One of the following error codes:

- missingQueryParameter: The URI is missing a required query-string parameter
- missingQueryValue: The URI is missing a required query-string parameter value
- invalidQuery: The query section of the URI is invalid.
- invalidBody: The request has an invalid body

7.1.1.4. **Type** Error401

Description: Unauthorized. (<https://tools.ietf.org/html/rfc7235#section-3.1>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error401Code	One of the following error codes: - missingCredentials: No credentials provided. - invalidCredentials: Provided credentials are invalid or expired

7.1.1.5. **enum** Error401Code

Description: One of the following error codes:

- missingCredentials: No credentials provided.
- invalidCredentials: Provided credentials are invalid or expired

7.1.1.6. **Type** Error403

Description: Forbidden. This code indicates that the server understood the request but refuses to authorize it. (<https://tools.ietf.org/html/rfc7231#section-6.5.3>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error403Code	This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes: - accessDenied: Access denied - forbiddenRequester: Forbidden requester - tooManyUsers: Too many users

7.1.1.7. **enum** Error403Code

Description: This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes:

- accessDenied: Access denied
- forbiddenRequester: Forbidden requester
- tooManyUsers: Too many users

7.1.1.8. Type Error404

Description: Resource for the requested path not found. (<https://tools.ietf.org/html/rfc7231#section-6.5.4>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	string	The following error code: - notFound: A current representation for the target resource not found
-------	--------	--

7.1.1.9. Type Error422

The response for HTTP status **422** is a list of elements that are structured using the **Error422** data type. Each list item describes a business validation problem. This type introduces the **propertyPath** attribute which points to the erroneous property of the request, so that the Buyer may fix it easier. It is highly recommended that this property should be used, yet remains optional because it might be hard to implement.

Description: Unprocessable entity due to a business validation problem. (<https://tools.ietf.org/html/rfc4918#section-11.2>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	Error422Code	One of the following error codes: - missingProperty: The property the Seller has expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the Seller system - unexpectedProperty: Additional property, not expected by the Seller has been provided - tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold. - otherIssue: Other problem was identified (detailed information provided in a reason)
-------	------------------------------	---

Name	Type	Description
propertyPath	string	A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer (https://tools.ietf.org/html/rfc6901).

7.1.1.10. **enum Error422Code**

Description: One of the following error codes:

- missingProperty: The property the Seller has expected is not present in the payload
- invalidValue: The property has an incorrect value
- invalidFormat: The property value does not comply with the expected value format
- referenceNotFound: The object referenced by the property cannot be identified in the Seller system
- unexpectedProperty: Additional property, not expected by the Seller has been provided
- tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold.
- otherIssue: Other problem was identified (detailed information provided in a reason)

7.1.1.11. **Type Error500**

Description: Internal Server Error. (<https://tools.ietf.org/html/rfc7231#section-6.6.1>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	string	The following error code: - internalError: Internal server error - the server encountered an unexpected condition that prevented it from fulfilling the request.
-------	--------	--

7.1.1.12. **Type Error501**

Description: Not Implemented. Used in case Seller is not supporting an optional operation (<https://tools.ietf.org/html/rfc7231#section-6.6.2>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	string	The following error code: - notImplemented: Method not supported by the server
-------	--------	--

7.2. Management API Data model

Figure 12 presents the whole Billing Management data model the data types, requirements related to them and mapping to MEF 134 specifications are discussed later in this section.

Name	Type	M/O	Description	MEF 134
billCycle	string	M	The identifier of the Billing Cycle iteration.	Invoicing Cycle Identifier
billDate	date-time format = date-time	M	Date the Bill was issued.	Invoice Date
billDocument	AttachmentURL	M	URL pointing to PDF file containing printable version of the Customer Bill.	Bill Document
billNo	string	M	A number assigned to the Bill by the Seller.	Invoice Number
billingPeriod	TimePeriod	M	The Start and Stop Dates of the Billing Period.	Invoicing Period
category	CustomerBillCategory	M	The type of Bill. One of the following: - normal - duplicate - trial	Invoice Category
credits	Money	M	Amount of credits included.	Credits
customerBillItem	CustomerBillItemRef[]	M	A reference to the Bill Items for this Bill.	Invoice Line Item
discounts	Money	M	Amount of discounts included.	Discounts
fees	Money	M	Amount of fees included.	Fees
financialAccount	FinancialAccountRef	M	A Financial Account within the Seller.	Financial Account
lastUpdate	date-time format = date-time	M	The date when the Bill was last modified.	Last Update Date

Name	Type	M/O	Description	MEF 134
paymentDueDate	date-time <i>format = date-time</i>	M	The date by which payment of the Amount Due must be received by the Seller.	Payment Due Date
runType	CustomerBillRunType	M	The Billing cycle as set by the Seller.	Run Type
relatedContactInformation	RelatedContactInformation[]	M	A party related to this Bill.	Related Contact Information
remainingAmount	Money	M	An amount of money that still requires payment.	Remaining Amount
state	CustomerBillStateType	M	The state of the Bill.	Invoice State
taxExcludedAmount	Money	M	The amount of money due without taxes being calculated for the current Billing Cycle.	Tax Excluded Amount
taxIncludedAmount	Money	M	The amount of money due with taxes being calculated for the current Billing Cycle.	Tax Included Amount
taxItem	TaxItem[]	M	A tax item is created for each tax rate and tax type used in the bill.	Tax Items

7.2.1.2. Type CustomerBill_Find

Description: A legal document generated by the Seller to the Buyer relating to charges associated to Products provided by the Seller to the Buyer.

Name	Type	M/O	Description	MEF 134
------	------	-----	-------------	---------

Name	Type	M/O	Description	MEF 134
id	string	M	An identifier assigned to the Bill by the Seller.	Invoice Identifier
billingAccount	BillingAccountRef	O	An identifier for the Billing Account that is unique within the Seller and is assigned by the Seller.	Invoicing Account
billNo	string	M	A number assigned to the Bill by the Seller.	Invoice Number
billingPeriod	TimePeriod	O	The Start and Stop Dates of the Billing Period.	Invoicing Period
category	CustomerBillCategory	M	The type of Bill. One of the following: - normal - duplicate - trial	Invoice Category
state	CustomerBillStateType	M	The state of the Bill.	Invoice State

7.2.1.3. Type CustomerBillItem

Description: One or more rows in a Bill that represent charges associated with a Product instance.

Name	Type	M/O	Description	MEF 134
id	string	M	The CustomerBillItem identifier.	Item Identifier
href	string	O	Reference of the CustomerBillItem.	Not represented in MEF 134
appliedTax	CustomerBillItemTax[]	M	Taxes associated with the Bill Item.	List of Taxes
appliedFee	CustomerBillItemFee[]	M		List of Fees
customerBillItemType	MEFPriceType	M	The type of charge related to the Bill Item.	Charge Type
description	string	M	A text description of the charge.	Charge Description
periodCoverage	TimePeriod	M	The start and end dates of Billing for the Bill Item.	Invoicing Start Date Invoicing End Date
product	ProductRef	M	The reference to the Product that is the subject of the Item.	Product Identifier

Name	Type	M/O	Description	MEF 134
productOrderItem	MEFProductOrderItemRef	M	Identifier of the POI with reference to the Product Order.	Product Order Identifier Product Order Item Reference Identifier
productName	string	M	The name of the Product that is the subject of the Bill Item.	Product Name
state	CustomerBillItemStateType	M	The state of the Bill Item.	Invoice Line Item State
taxExcludedAmount	Money	M	The amount of money of the charge before taxes and fees are calculated and applied	Duty Free Amount
unit	string	M	The rate per unit for the Bill determined during or after the Billing Process.	Not represented in MEF 134
unitRate	Money	M	The rate per unit for the Bill determined during or after the Billing Process.	Unit Rate
unitQuantity	number	M	The number of units.	Unit Quantity

7.2.1.4. enum CustomerBillCategory

Description: The type of Bill.

Value	MEF 134	Description
normal	NORMAL	An Bill for the Billing Cycle
duplicate	DUPLICATE	A copy of an Bill that has been provided
trial	TRIAL	An Bill that is sent by the Seller to the Buyer for test Billing purposes to assist in assuring that their Billing implementations are compatible. The Trial Bill is not paid.

7.2.1.5. Type CustomerBillItemFee

Description: Fees associated with the Bill Item.

Name	Type	M/O	Description	MEF 134
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Name	Type	M/O	Description	MEF 134
category	CustomerBillItemFeeCategory	O	The category of the Fee. One of the following: - recurring - nonRecurring - other	Fee Category
description	string	O	A description of the type of Fee.	Fee Description
rate	float format = float	O	The rate at which the Fee is calculated.	Fee Rate
amount	Money	O	The amount of money of the Fee.	Fee Amount

7.2.1.6. **enum** CustomerBillItemFeeCategory

Description: The category of the Fee. One of the following:

- recurring
- nonRecurring
- other

7.2.1.7. Type CustomerBillItemRef

Description: A reference to a Customer Bill resource.

Name	Type	M/O	Description	MEF 134
href	string	O	Hyperlink to the referenced Customer Bill.	Not represented in MEF 134
id	string	M	Identifier of the referenced Customer Bill.	Item Identifier

7.2.1.8. **enum** CustomerBillItemStateType

Description: The state of the Bill Item.

state	MEF 134 name	Description
credit	CREDIT	The Seller has agreed with the Buyer that disputed charges are in error and has provided a credit for the amount in error.
disputeBeingInvestigated	DISPUTE_BEING_INVESTIGATED	A Buyer has disputed charges included in a Bill and those charges are in the Sellers Dispute Process. Disputes cannot exit the dispute process until the Dispute is resolved.

state	MEF 134 name	Description
generated	GENERATED	A Bill that has been created based on the completion of a Billing Cycle and a Bill Notification has been sent to the Buyer.
paymentDue	PAYMENT_DUE	The Seller has completed their dispute process and they are expecting payment from the Buyer.
settled	SETTLED	A Bill for which all payment due has been received or for which payment is due.
withdrawn	WITHDRAWN	A Seller has agreed with the Buyer that disputed charges are in error and has withdrawn the Bill.

7.2.1.9. Type CustomerBillItemTax

Description: The applied billing tax rate represents the taxes applied billing rate it refers to. It is calculated during the billing process.

Name	Type	M/O	Description	MEF 134
category	CustomerBillItemTaxCategory	O	The category of the Tax.	Tax Category
description	string	O	A description of the type of Tax.	Tax Description
rate	float format = float	O	The rate at which the Tax is calculated.	Tax Rate
amount	Money	O	The amount of money of the Tax.	Tax Amount

7.2.1.10. enum CustomerBillItemTaxCategory

Description: The category of the Tax. One of the following:

- country
- state
- county
- city
- other

7.2.1.11. enum CustomerBillRunType

Description: The Billing cycle as set by the Seller.

Value	MEF 134	Description
onCycle	On_Cycle	Bill created as a result of the normal Invoicing Cycle
offCycle	Off_Cycle	Bill created outside the normal Invoicing Cycle. This may be based on a Buyer request or for other reasons.

7.2.1.12. enum CustomerBillStateType

Description: The state of the Bill.

state	MEF 134 name	Description
generated	GENERATED	A Bill that has been created based on the completion of an Billing Cycle and an Bill Notification has been sent to the Buyer.
paymentDue	PAYMENT_DUE	The Seller has completed their dispute process and they are expecting payment from the Buyer
settled	SETTLED	A Bill for which all payment due has been received or for which payment is due.

7.2.1.13. Type FinancialAccountRef

Description: A Financial Account within the Seller.

Name	Type	M/O	Description	MEF 134
id	string	M	An identifier for the Financial Account that is unique within the Seller and is assigned by the Seller.	Identifier
href	string	O	Unique reference of the account	Not represented in MEF 134
name	string	O	The name of the Financial Account.	Name
type	string	O	The type of the Financial Account.	Type

7.2.1.14. enum MEFPriceType

Description: The type of charge related to the Bill Item.

Value	MEF 134
recurring	RECURRING
nonRecurring	NON_RECURRING
usageBased	USAGE_BASED

7.2.1.15. Type MEFProductOrderItemRef

Description: It's a ProductOrder item

Name	Type	M/O	Description	MEF 134
------	------	-----	-------------	---------

Name	Type	M/O	Description	MEF 134
productOrderHref	string	O	Reference of the related ProductOrder.	Not represented in MEF 134
productOrderId	string	M	Unique identifier of a ProductOrder.	Product Order Identifier
productOrderItemId	string	M	Id of an Item within the Product Order	Product Order Item Reference Identifier

7.2.1.16. Type PaymentItem

Description: A payment that has been received.

Name	Type	M/O	Description	MEF 134
id	string	M	An identifier for the payment that is unique within the Buyer Billing Account and is assigned by the Seller.	Payment Identifier
amount	Money	O	The amount of money received.	Amount
paymentMethod	PaymentMethod	O	The specific means of payment.	Payment Method
paymentDate	date-time format = date-time	O	The Date the payment was received.	Payment Date

7.2.2. Common

Types described in this subsection are shared among two or more Cantata and Sonata APIs.

7.2.2.1. Type AppliedPayment

Description: A list of details of a payment that has been received from the Buyer.

Name	Type	M/O	Description	MEF 134
appliedAmount	Money	O	The amount of money that was received from the Buyer and applied to the Bill as payment.	Applied Amount
payment	PaymentItem	O	A list of payment items that have been received.	Payment Item

7.2.2.2. Type AttachmentURL

Description: The URL pointing to an Attachment for download.

Name	Type	M/O	Description	MEF 134
url	string	O	The URL pointing to an Attachment for download.	Bill Document

7.2.2.3. Type BillingAccountRef

Description: An identifier for the Billing Account that is unique within the Seller

Name	Type	M/O	Description	MEF 134
id	string	M	Unique-Identifier	Invoicing Account Identifier

7.2.2.4. Type FinancialAccountRef

Description: A Financial Account within the Seller.

Name	Type	M/O	Description	MEF 134
id	string	M	An identifier for the Financial Account that is unique within the Seller and is assigned by the Seller.	Identifier
href	string	O	Unique reference of the account	Not represented in MEF 134
name	string	O	The name of the Financial Account.	Name
type	string	O	The type of the Financial Account.	Type

7.2.2.5. Type FieldedAddressRepresentation

Description: A type of Address that has a discrete field and value for each type of boundary or identifier down to the lowest level of detail. For example "street number" is one field, "street name" is another field, etc.

Name	Type	M/O	Description	MEF 134
streetNr	string	O	Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.	Street Number
streetNrSuffix	string	O	The first street number suffix (in a street number range) or the suffix for the street number if there is no range	Street Number Suffix
streetNrLast	string	O	Last number in a range of street numbers allocated to an Address	Street Number Last
streetNrLastSuffix	string	O	Last street number suffix for a ranged Address	Street Number Last Suffix
streetPreDirection	string	O	The direction of the street that appears before the Street Name	Street Pre-Direction
streetName	string	O	Name of the street or other street type	Street Name

Name	Type	M/O	Description	MEF 134
streetType	string	O	The type of street (e.g., alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf)	Street Type
streetPostDirection	string	O	A modifier denoting a relative direction that appears after the Street Name.	Street Post-Direction
poBox	string	O	Number identifying a specific location in a post office.	PO Box Number
locality	string	O	An area of defined or undefined boundaries within a local authority or other legislatively defined area.	Locality
city	string	O	City in which the Address is located.	City
postcode	string	O	A descriptor for a postal delivery area used to speed and simplify the delivery of mail (also known as zip code)	Postal Code
postcodeExtension	string	O	The extension used on a postal code. Note: there are different use codes for this attribute depending upon the country.	Postal Code Extension
stateOrProvince	string	O	The State or Province in which the Address is located.	State or Province
countryCode	string minLength = 2 maxLength = 2	O	Country in which the Address is located, defined using two characters as defined in ISO 3166	Country
subUnit	SubUnit[]	O	The Sub Unit represented as a list. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A	Sub Units
buildingName	string	O	The well-known name of a building that is located at this Address (e.g., where there is one Address for a campus).	Building Name
privateStreetNumber	string	O	Street number on a private street within the Address.	Private Street Number
privateStreetName	string	O	Private streets internal to a property (e.g., a university) may have internal names that are not recorded by the land title office.	Private Street Name
language	string minLength = 2 maxLength = 2	O	The language in which the address is expressed. It MUST use the ISO 639:2023 two letter code 639:2023	Language

7.2.2.6. Type SubUnit

Description: Allows for sub unit identification

Name	Type	M/O	Description	MEF 134
subUnitNumber	string	M	The discriminator used for the subunit, often just a simple number but may also be a range.	Sub Unit Name
subUnitType	string	M	The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.	Sub Unit Type

7.2.2.7. Type Money

Description: A base value business entity used to represent money

Name	Type	M/O	Description	MEF 134
unit	string	M	Currency (ISO4217 norm uses 3 letters to define the currency)	Currency
value	float format = float	M	A positive floating point number	Value

7.2.2.8. **enum** PaymentMethod

Description: The specific means of payment.

Value	MEF 134
check	CHECK
wireTransfer	WIRE_TRANSFER
electronic	ELECTRONIC
cash	CASH
other	OTHER

7.2.2.9. Type ProductRef

Description:

Name	Type	M/O	Description	MEF 134
id	string	M	Unique identifier of a related entity.	Product Identifier
href	string	O	Reference of the related entity.	Not represented in MEF 134

7.2.2.10. Type RelatedContactInformation

Description: Contact data for a person or organization that is involved in the product offering qualification. In a given context it is always specified by the Seller (e.g. Seller Contact

Information) or by the Buyer.

Name	Type	M/O	Description	MEF 134
role	string	M	A role of the particular contact in the request	Not represented in MEF 134
number	string	M	Phone number	Contract Phone Number
emailAddress	string	M	Email address	Contact email Address
postalAddress	FieldedAddressRepresentation	O	Identifies the postal address of the person or office to be contacted.	Contact Postal Address
organization	string	O	The organization or company that the contact belongs to	Contact Organization
name	string	M	Name of the contact	Contact Name
numberExtension	string	O	Phone number extension	Contract Phone Number Extension

7.2.2.11. Type TaxItem

Description: A tax item is created for each tax rate and tax type used in the bill.

Name	Type	M/O	Description	MEF 134
taxCategory	string	O	The Tax Category for this tax item.	Tax Category
taxRate	float format = float	O	The Tax Rate for this Tax Item.	Tax Rate
taxAmount	Money	O	The amount of money calculated for this Tax Item.	Tax Amount

7.2.2.12. Type TimePeriod

Description: A period of time, either as a deadline (endDateTime only) a startDateTime only, or both

Name	Type	M/O	Description	MEF 134
endDateTime	date-time format = date-time	O	The date the Billing Period ended.	End Date
startDateTime	date-time format = date-time	O	The date the Billing Period started.	Start Date

7.2.3. Notification registration

Notification registration and management are done through `/hub` API endpoint. The below sections describe data models related to this endpoint.

7.2.3.1. Type EventSubscription

Description: Sets the communication endpoint address the service instance must use to deliver notification information

Name	Type	M/O	Description	MEF 134
id	string	M	Id of the listener	Not represented in MEF 134
callback	string	M	The callback being registered.	Return Address Information
query	string	O	additional data to be passed	Notification Type

7.2.3.2. Type EventSubscriptionInput

Description: Sets the communication endpoint address the service instance must use to deliver notification information

Name	Type	M/O	Description
callback	string	M	This callback value must be set to <code>*host*</code> property from Buyer Notification property is appended with the base path and notification resource path specific notification is sent. E.g. for "callback": "http://buyer.com/listenerEndpoint", is sent <code>'http://buyer.com/listenerEndpoint/mefApi/sonata/customerBillManagement/v2'</code>
query	string	O	This attribute is used to define to which type of events to register customerBillCreateEvent". To subscribe for more than one event type, <code>'eventType=customerBillCreateEvent,customerBillStateChangeEvent'</code> . The <code>eventType</code> is defined in <code>'CustomerBillEventType'</code> in <code>billingNotification.api.yaml</code> . An empty query is 1 subscription for all event types.

7.3. Notification API Data Model

Figure 13 presents the Billing Notification data model the data types, requirements related to them and mapping to MEF 134 specifications are discussed later in this section.

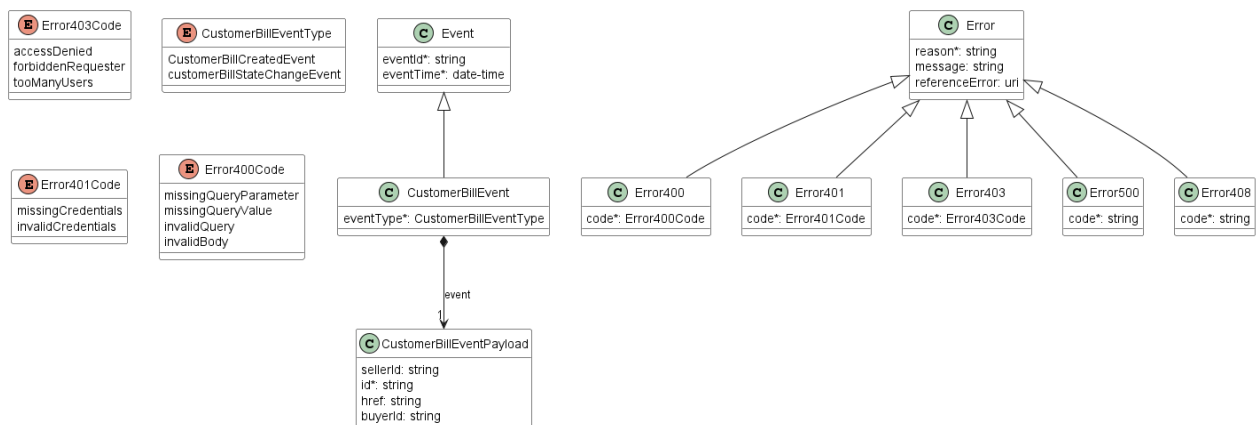


Figure 13. Billing Notification Data Model

7.3.1. Type Event

Description: Event class is used to describe information structure used for notification.

Name	Type	M/O	Description	MEF 134
eventId	string	M	Id of the event	Not represented in MEF 134
eventTime	date-time format = date-time	M	Date-time when the event occurred	Not represented in MEF 134
eventType	string	M	The type of the notification.	Notification Type
event	object	M	The event linked to the involved resource object	Invoice Identifier

7.3.2. Type CustomerBillCreateEvent

Description:

Inherits from:

- [Event](#)

Name	Type	M/O	Description	MEF 134
eventType	string	M	Indicates the type of the event.	Notification Type
event	CustomerBillCreateEventPayload	M	A reference to the object that is source of the notification.	Invoice Identifier

7.3.3. Type CustomerBillCreateEventPayload

Description: The identifier of the Bill being subject of this event.

Name	Type	M/O	Description	MEF 134
id	string	M	ID of the Bill attributed by quoting system	Invoice Identifier
href	string	O	Hyperlink to access the Bill	Not represented in MEF 134

7.3.4. Type CustomerBillStateChangeEvent

Description:

Inherits from:

- [Event](#)

Name	Type	M/O	Description	MEF 134
eventType	string	M	Indicates the type of the event.	Notification Type
event	CustomerBillStateChangeEventPayload	M	A reference to the object that is source of the notification.	Invoice Identifier

7.3.5. Type CustomerBillStateChangeEventPayload

Description: The identifier of the Bill being subject of this event.

Name	Type	M/O	Description	MEF 134
id	string	M	ID of the Bill attributed by quoting system	Invoice Identifier
href	string	O	Hyperlink to access the Bill	Not represented in MEF 134
state	CustomerBillStateType	M	The state of the Bill.	Not represented in MEF 134

8. References

- [ISO4217](#) International Standards Organization ISO 4217:2015, 2015
- [JSONSchema](#) JSON Schema main page
- [MEF 55.1](#), Lifecycle Service Orchestration (LSO): Reference Architecture and Framework, February 2021
- [MEF 55.1.1](#), Amendment to MEF 55.1: Reference Architecture and Framework - Terminology, June 2023
- [MEF 128.1](#), LSO API Security Profile, April 2024
- [MEF 134](#), Billing and Invoice Business Requirements and Use Cases, February 2023
- [Mplify 150](#), Installation Place and Service Site Management Business Requirements and Use Cases, June 2025
- [OAS-v3](#), February 2020
- [REST](#) Fielding, Roy Thomas, Architectural Styles and the Design of Network-based Software Architectures (Ph.D.).
- [RFC 2119](#), Key words for use in RFCs to Indicate Requirement Levels, March 1997
- [RFC 3986](#) Uniform Resource Identifier (URI): Generic Syntax, January 2005
- [RFC 7231](#), Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content, June 2014
- [RFC 8174](#), Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words, May 2017
- [TMF630](#), TMF630 API Design Guidelines 4.2.0
- [TMF678](#), TMF678 Customer Bill Management API User Guide v4.0.1 Doc Checklist:

Appendix A Acknowledgments

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